

CO1_AT3

Data Independence its Significance in DBMS Architecture

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Introduction

What is a Database Management System (DBMS)?

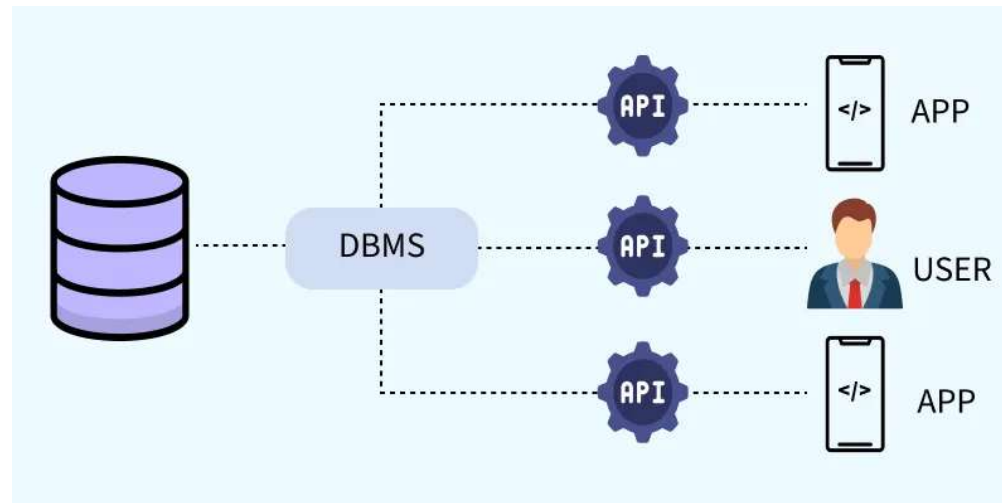
- A Database Management System (DBMS) is software that allows users to create, store, retrieve, update, and manage data efficiently. It acts as an interface between users/applications and the database.

Examples include:

- MySQL
- Oracle Database
- Microsoft SQL Server
- PostgreSQL
- MongoDB

DBMS ensures:

- Data security
- Data integrity
- Efficient data retrieval



Objectives

By the end of this presentation, you will understand:

- DBMS Architecture
- Three Schema Architecture
- Meaning of Data Independence
- Types of Data Independence
- Logical Data Independence
- Physical Data Independence
- Importance in modern databases
- Real-world examples



DBMS Architecture

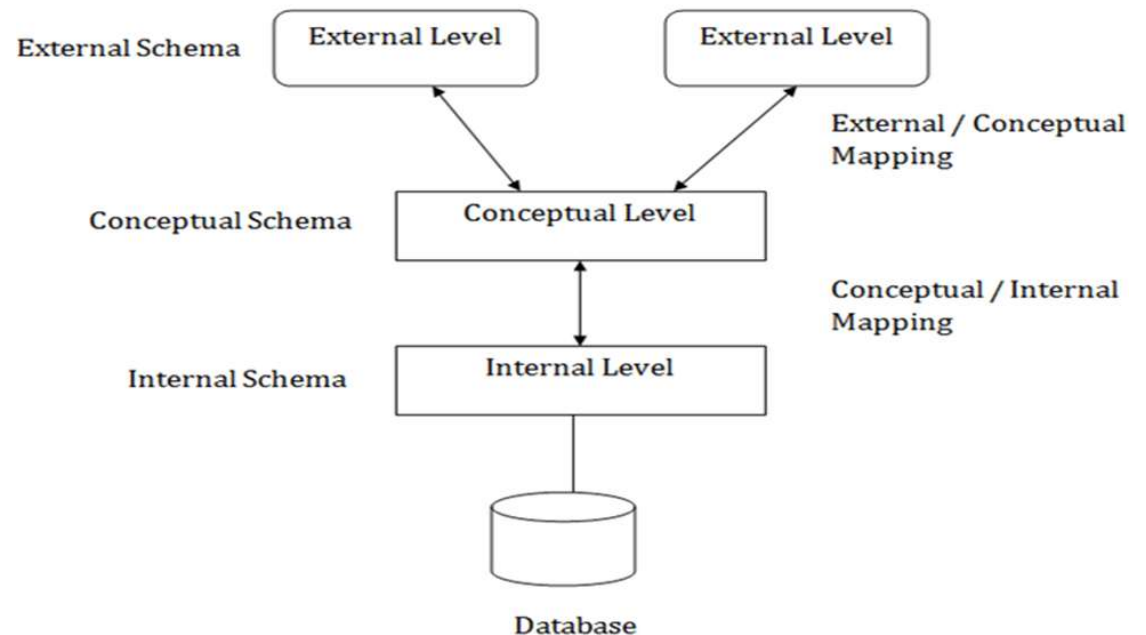
- DBMS Architecture is a framework that defines how data is stored, organized, and accessed.

It separates:

- User Interface
- Database Logic
- Physical Storage

This separation makes databases:

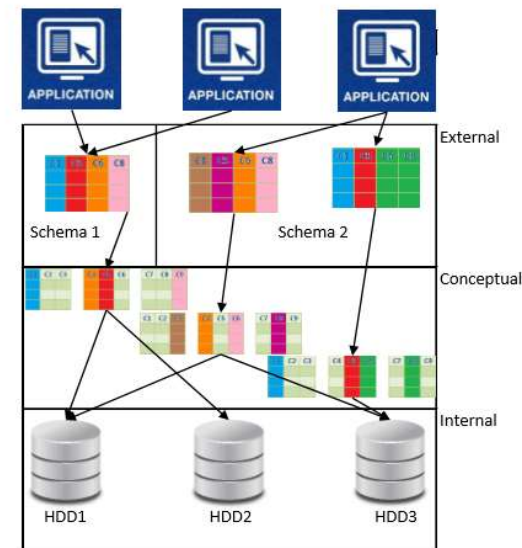
- Secure
- Flexible
- Easy to maintain



ANSI/SPARC Three Schema Architecture

The ANSI/SPARC Architecture divides the database into three levels.

- External Level
- Conceptual Level
- Internal Level



This architecture provides Data Independence.

External Level

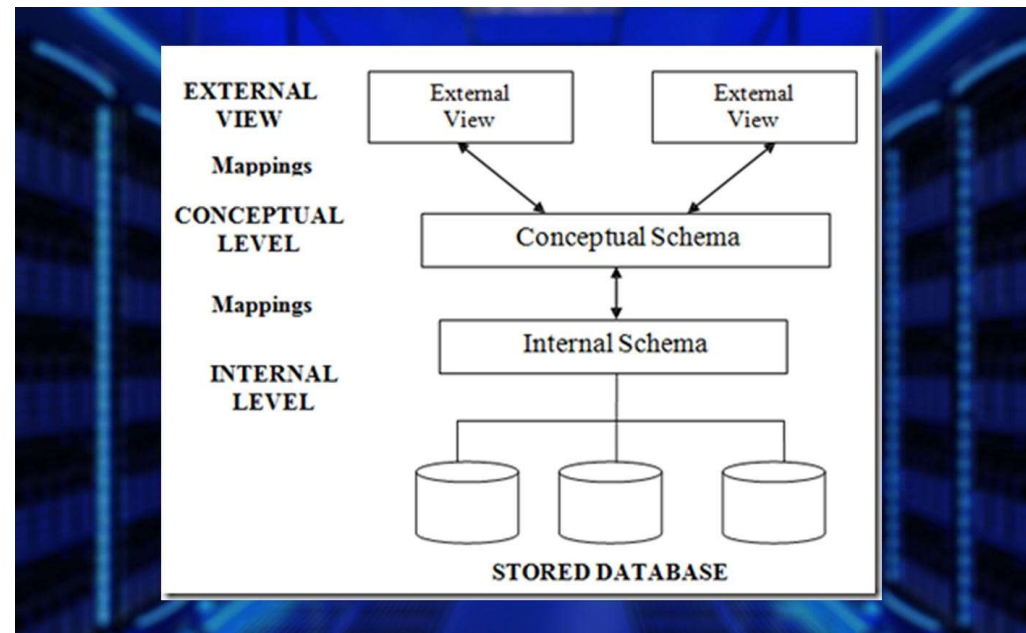
- External Level is the user's view of the database.

Features:

- Different users see different data.
- Hides unnecessary information.
- Improves security.
- Multiple external views are possible.

Example:

- Students can only see marks.
- Teachers can update marks.
- Principal can view everything.

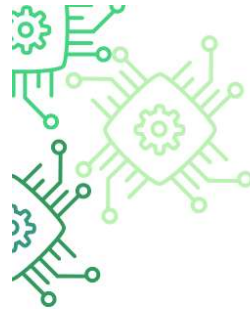


Conceptual Level

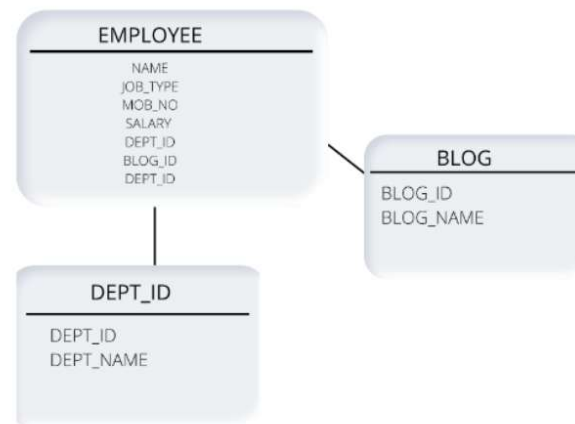
This level describes the complete logical structure of the database.

Includes:

- Tables
- Relationships
- Constraints
- Data Types



Database Schema



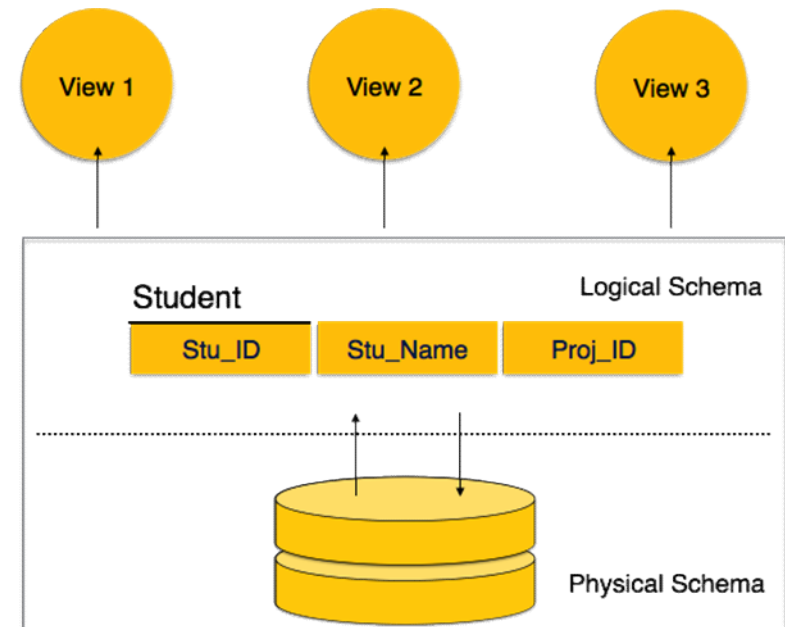
The users cannot see physical storage.

Internal Level

Internal Level explains how data is physically stored.

Includes:

- Files
- Indexes
- Blocks
- Memory Allocation
- Compression
- This level is managed by the DBMS itself.

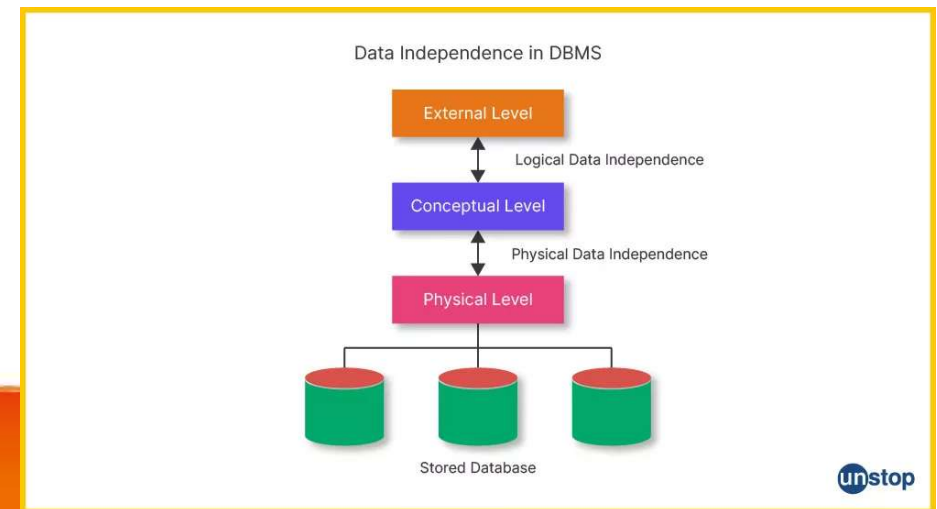


What is Data Independence?

Definition:

- Data Independence is the ability to modify the database schema at one level without affecting the schema at the next higher level.

In simple words, Changes inside the database should not require rewriting application programs.



Why is Data Independence Important?

Benefits:

- Easier maintenance
 - Better flexibility
 - Lower software cost
 - Improved scalability
 - Longer database life
 - Faster upgrades
 - Reduced application failures
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Types of Data Independence

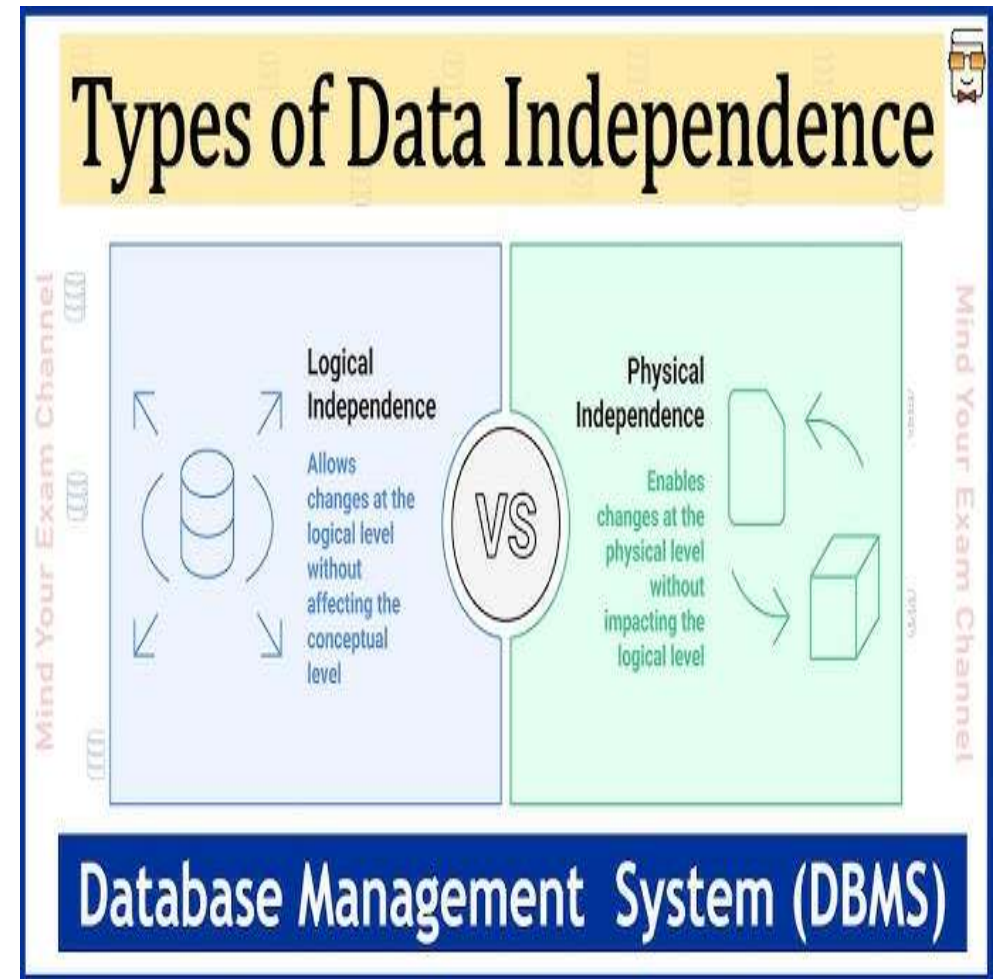
There are two types:

- 1. Physical Data Independence:

Changes in physical storage do not affect logical structure.

2. Logical Data Independence

Changes in logical structure do not affect user applications.



Physical Data Independence

Definition:

- The ability to modify the internal schema without changing the conceptual schema.
- Storage changes remain invisible to users.

Examples:

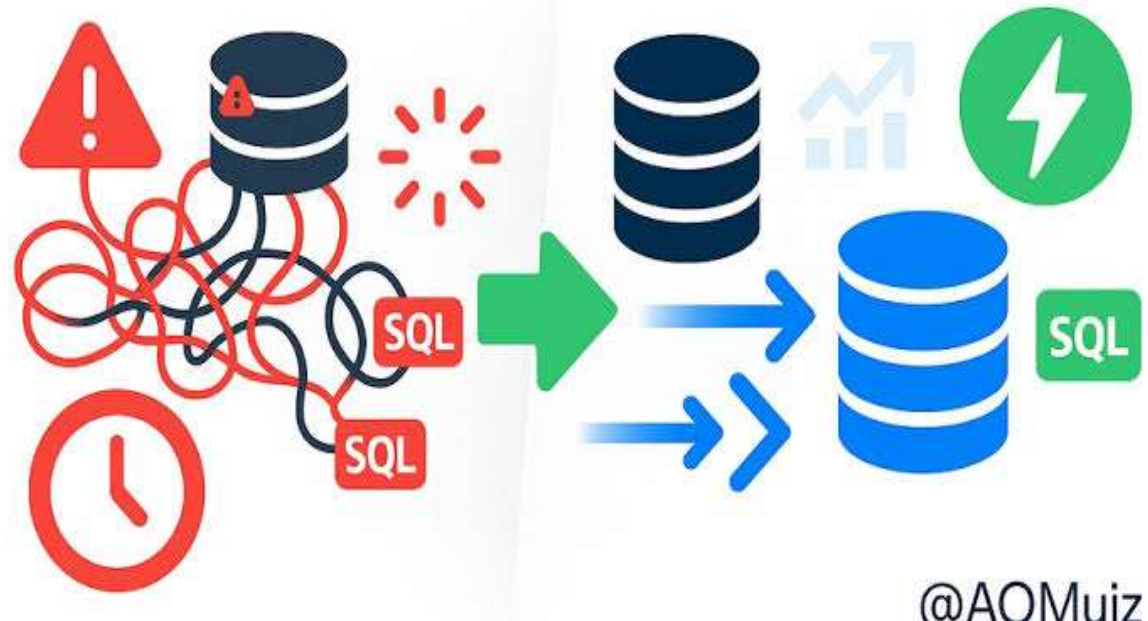
- Creating indexes
 - Changing file organization
 - Data compression
 - SSD upgrade
 - Partitioning
- 

Advantages of Physical Data Independence

Advantages:

- Better Performance
- Easy Hardware Upgrade
- Faster Query Processing
- Lower Maintenance Cost
- No Application Changes
- Efficient Storage Management

From Craggish to Lightning Fast
Database Performance Optimization



@AOMuiz

Example of Physical Data Independence

Example:

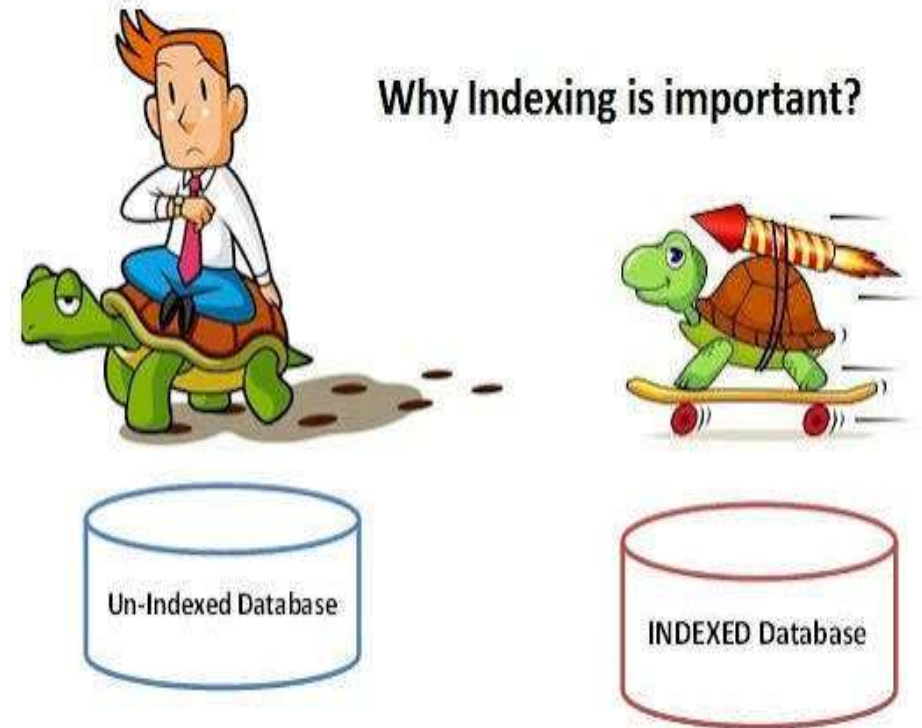
- A company changes storage from HDD to SSD.

OR

- Creates an index to speed up searching.

Result:

- Applications continue working exactly the same.
- Only performance improves



Logical Data Independence

Definition:

- Logical Data Independence means changes to the conceptual schema do not affect external views.
- Users continue using applications without modification.

Examples:

- Add new columns
- Add new tables
- Merge tables
- Split tables

Data dependent

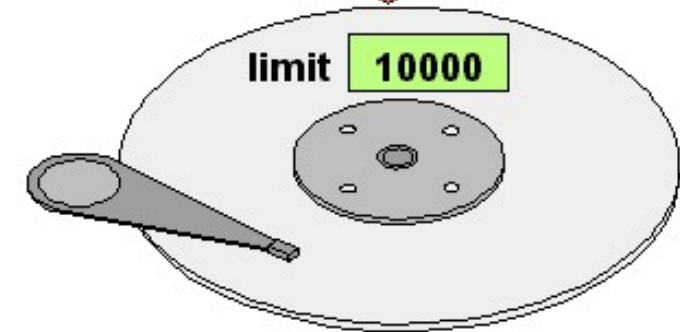
If credit_limit > 10000

1

Data independent

If credit_limit > limit

2



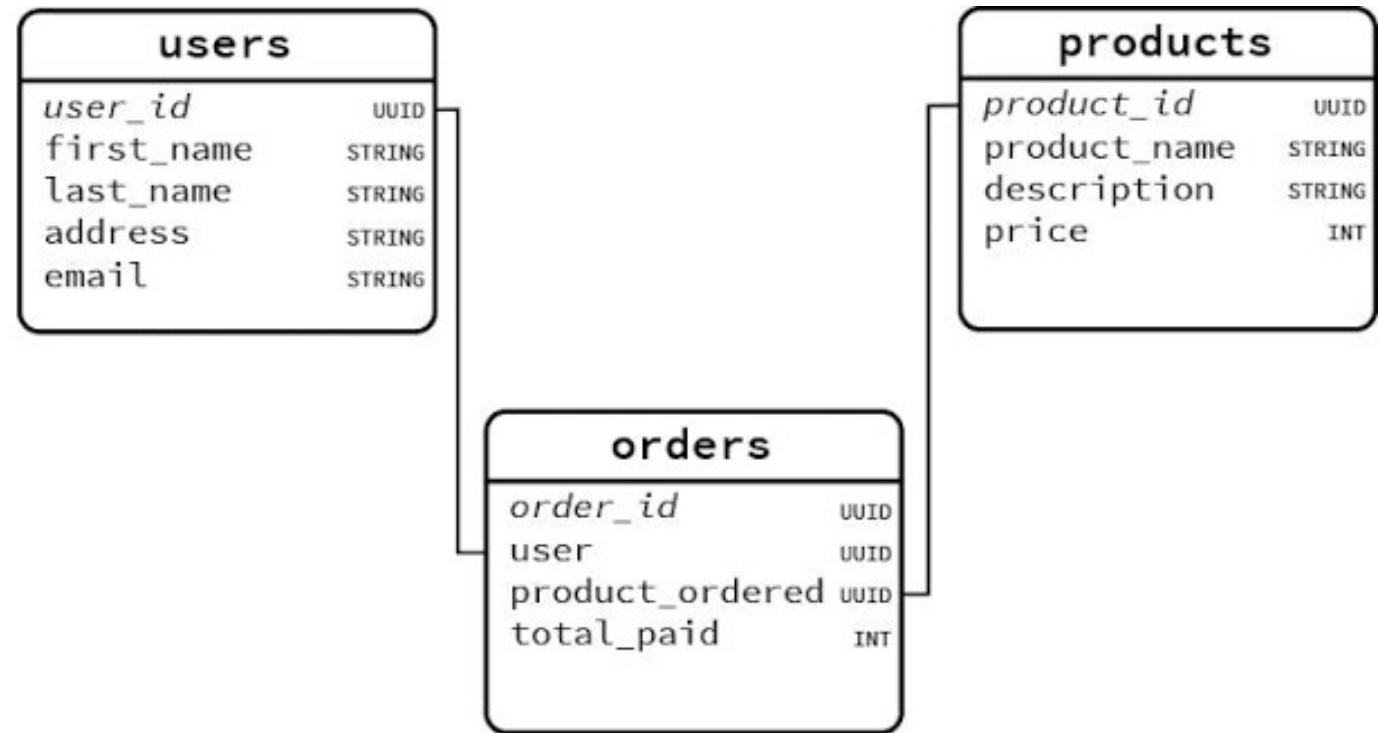
Advantages of Logical Data Independence

Advantages:

- Easier database evolution
 - Better flexibility
 - Supports business growth
 - Reduces application redesign
 - Easy addition of new features
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Example of Logical Data Independence

- Old Student Table
- StudentID
- Name
- Department
- New Student Table
- StudentID
- Name
- Department
- Email
- Phone Number
- The existing application still works.



Logical vs Physical Data Independence

Comparison between conceptual, logical, physical data models

This slide represents the difference between main models of data modeling, such as conceptual, logical, and physical. The comparison is based on features covering entity names, relationships, attributes, primary and foreign keys, and so on.

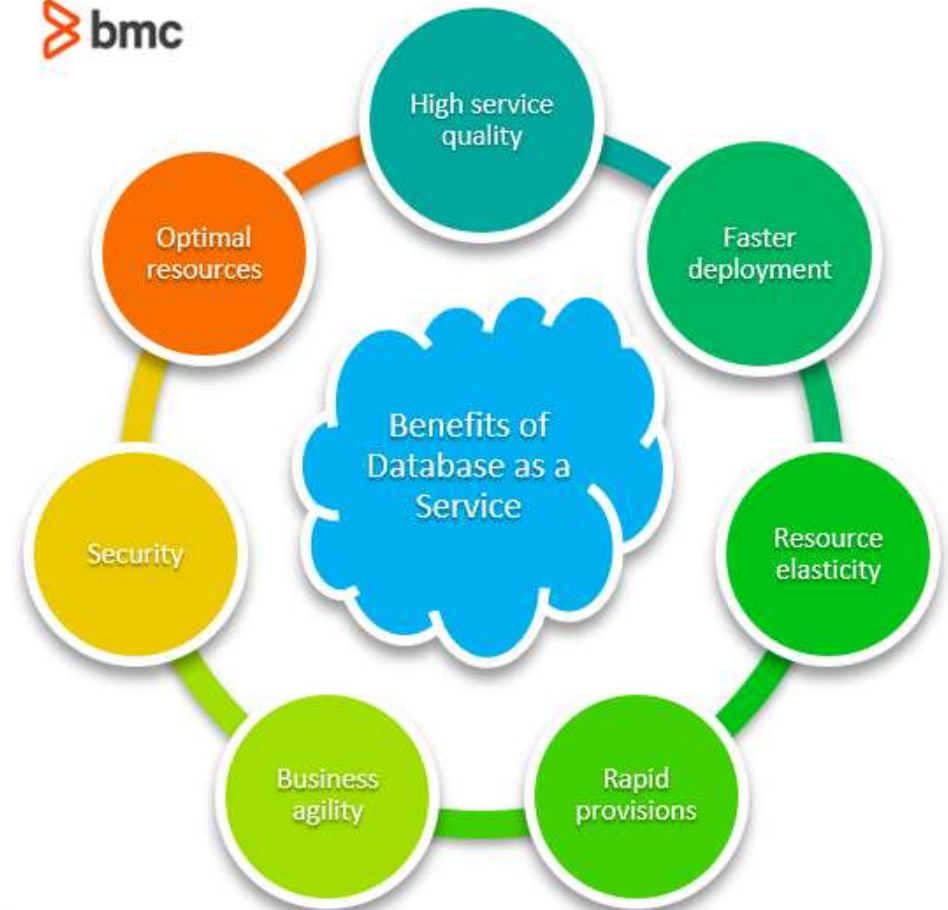
Feature	Conceptual data model	Logical data model	Physical data model
Entity names	✓	✓	
Entity relationships	✓	✓	
Attributes		✓	
Primary keys		✓	✓
Foreign keys		✓	✓
Naming convention	No restrictions	No restrictions	Naming conventions governed by DBMS
Owner column data type	Business stakeholders	Data architects & business analysts	Database administrator & developers
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Significance in DBMS Architecture

Data Independence helps:

- Database scalability
- Easy maintenance
- Data security
- System reliability
- Better application development
- Reduced downtime
- Technology upgrades



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Thank you

